

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 2– Case Study (2 QUESTIONS)

Course Code:	ITA05	Course Title:	COMPUTER VISION
CO Assessed:	CO3 – Precision (BL4)	Assessment:	Assessment Tool 1 – Case Study
Weightage:	40% of CIA	Total Marks:	20 (2 Questions × 10Marks)
CO1 Statement:	<i>Analyze and extract image features using detection and matching techniques for effective image representation and comparison..(BL4)</i>		
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Section / Batch:	2024-slot D	Date:	10.8.2024

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Code of Conduct

I T.Chaitra(192425217) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student:T.Chaitra

Instructions

- This test contains 2 questions, each carrying 10 marks. Total: 20 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 60 minutes.

Section / Topic	Focus	Q Nos	Marks
Feature Detection and Matching	Preprocessing, Edge Detection, Harris Corner Detection, Hough Transform, SIFT	1	10
Feature Detection and Matching	PCA, Image Representation, Feature Matching, Image Processing Applications	2	10
GRAND TOTAL:			20 Marks

Annexure A –Case Study

1.A remote sensing application is used to align satellite images of the same geographical region captured at different times for monitoring environmental changes. The system initially relies on edge detection to identify matching regions between the images; however, due to differences in illumination, image orientation, and background details, the matching process becomes unreliable and produces inaccurate alignment results. The development team decides to replace the edge-based approach with Harris Corner Detection to identify stable and distinctive feature points. Analyze the limitations of edge detection in this application and explain how Harris Corner Detection improves feature localization, matching accuracy, and robustness for image registration under varying imaging conditions.

2.An autonomous surveillance system continuously captures high-resolution color images to detect vehicles and pedestrians in real time. Processing the original color images directly increases computational complexity and slows down feature extraction, making it difficult to achieve the required processing speed. To improve system performance, the engineers consider converting the images into grayscale or binary representations before applying feature detection algorithms. Analyze the impact of different image representations on computational efficiency, edge detection, and feature extraction performance, and justify the most appropriate image representation for achieving accurate and reliable object detection in real-time computer vision applications.

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding ; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

Faculty In-charge

Signature: _____

Date: _____

1. Harris Corner Detection for Satellite Image Registration

Introduction

In remote sensing, satellite images of the same geographical region are often captured at different times to monitor changes such as **deforestation, urban growth, floods, crop development, and land-use changes**. Before comparing these images, they must be accurately aligned. This process is called **image registration**.

Initially, the system uses **edge detection** to identify corresponding regions. However, edges are often unreliable because they can change due to illumination, orientation, noise, and background variations. **Harris Corner Detection** provides more stable feature points that can be used for accurate registration.

Limitations of Edge Detection

Edge detection identifies locations where there is a sudden change in image intensity.

For example:

- Buildings produce strong edges.
- Roads produce long edges.
- Forest boundaries produce edges.
- Mountains and geographical boundaries may also produce edges.

However, using edges alone for satellite image registration has several problems.

1. Sensitivity to illumination

Images captured at different times may have different sunlight conditions. Shadows and brightness changes can modify the appearance of edges.

2. Edges are not highly distinctive

An edge only represents a boundary. Many different locations can have similar edge structures.

For example, several buildings may have parallel edges. It becomes difficult to determine which edge in one image corresponds to an edge in another image.

3. Poor localization

An edge usually contains many pixels along a continuous line. Therefore, selecting one exact point from an edge for matching is difficult.

4. Background variations

Trees, clouds, shadows, water, and other background objects can produce additional edges, causing false matches.

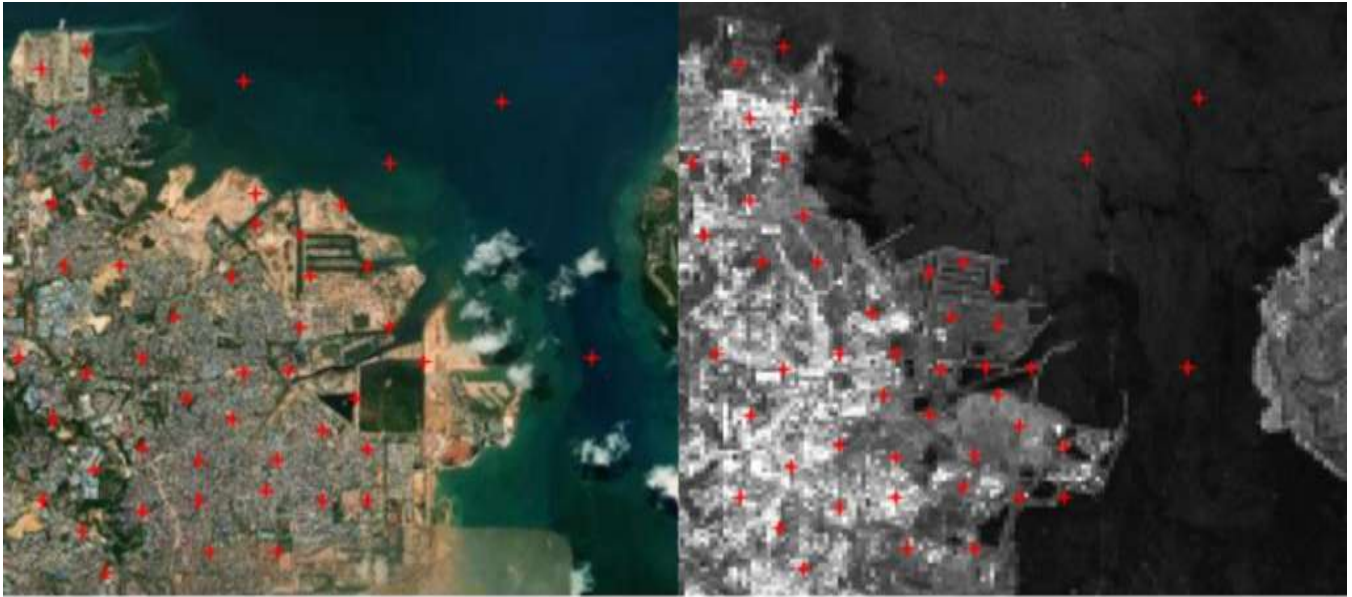
5. Orientation changes

If the satellite images are rotated or geometrically transformed, the appearance and position of edges may change significantly.

Harris Corner Detection

Harris Corner Detection identifies **corners**, which are points where the image intensity changes significantly in **two different directions**.

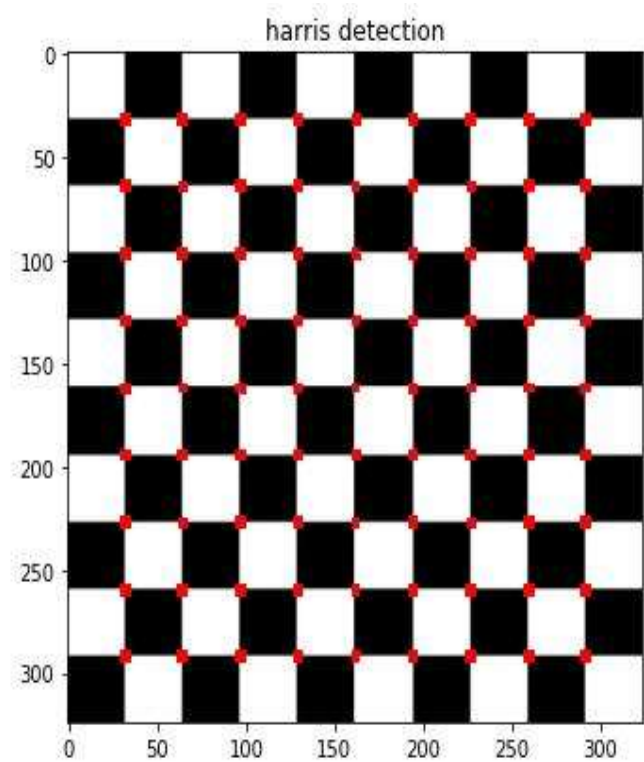
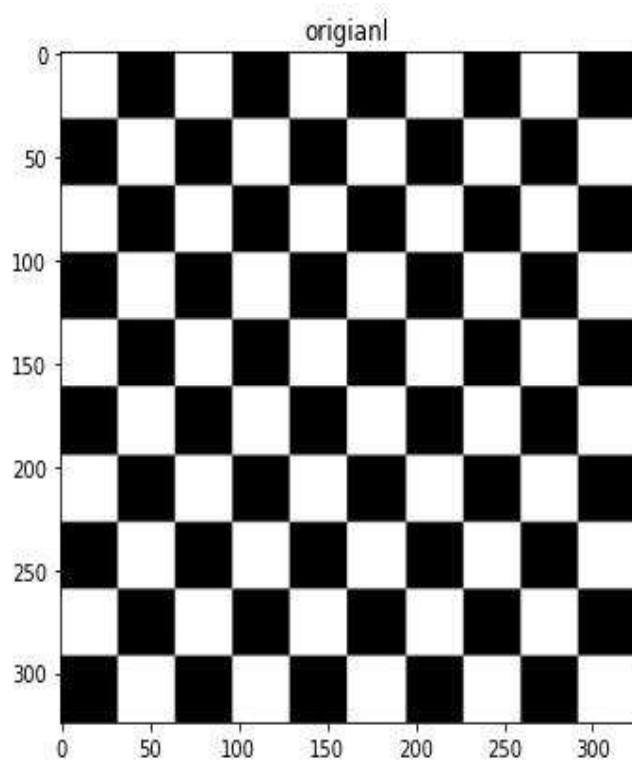
For example, the corner of a building contains strong intensity changes both horizontally and vertically.



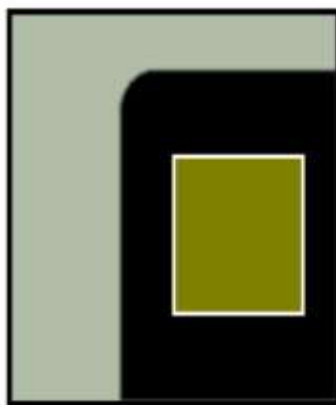
(a) Feature points detection by multi-scale Harris



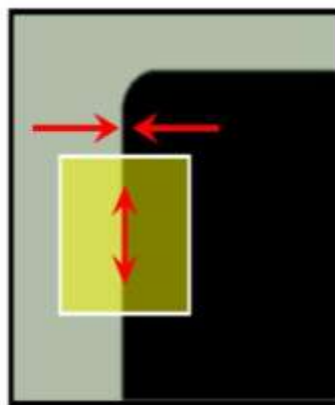
(b) Feature points detection by PC-Harris



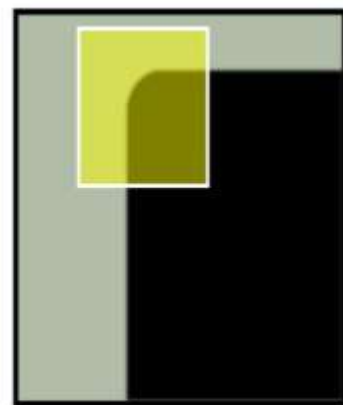
Corner Detection: Basic idea



"flat" region:
no change in
all directions



"edge" : no change
along the edge
direction



"corner" : significant
change in all directions
with small shift

The basic idea is:

Image → Grayscale → Calculate gradients → Structure matrix → Harris response → Select strong corners

The Harris response is commonly calculated using:
where:

- = structure tensor / second-moment matrix
- = determinant of the matrix
- = trace of the matrix
- = Harris constant, generally around 0.04–0.06

A large positive value of indicates a strong corner.

How Harris Improves Image Registration

1. Better Feature Localization

Corners provide **specific points** rather than long continuous edges.

For example, the corner of a building can be represented by one precise coordinate:

These coordinates can be compared between two satellite images.

Therefore, Harris provides more precise feature localization.

2. Better Matching Accuracy

Corners are more distinctive than simple edges.

Consider a building:

Edge-based:

Corner-based:



The corner contains information about intensity changes in multiple directions.

Therefore, the system can more reliably determine that a corner in Image 1 corresponds to the same geographical location in Image 2.

3. Robustness to Illumination Changes

Satellite images taken at different times may have different brightness levels.

Harris uses local intensity gradients rather than simply comparing the absolute brightness of pixels.

Therefore, important structural features such as:

- building corners
- road intersections
- field boundaries
- structural junctions

can remain useful even when illumination changes.

4. Robustness to Background Details

Background objects can generate many unwanted edges. Harris focuses on points where intensity changes occur in multiple directions.

This reduces the number of ambiguous features and improves matching.

5. Useful for Geometric Transformation

Satellite images may have:

- translation
- rotation
- scaling
- slight perspective differences

Corner points provide strong reference locations for estimating the geometric transformation between images.

For example:

where T represents the transformation required to align the images.

6. Improved Registration

After detecting Harris corners in both images, corresponding points can be matched.

The transformation can then be estimated using methods such as:

- Affine transformation
- Homography
- RANSAC

Finally, one image is warped so that both images overlap correctly.

Comparison

Feature	Edge Detection	Harris Corner Detection
Detects	Boundaries/edges	Distinctive corner points
Localization	Less precise	More precise
Matching	Difficult for repetitive edges	Easier using distinctive points
Illumination robustness	Relatively low	Better
Background sensitivity	High	Lower
Feature uniqueness	Low to moderate	High
Registration suitability	Limited	Very suitable

Conclusion

Edge detection is useful for identifying boundaries, but it is not ideal as the primary feature for satellite image registration because edges are often **long, repetitive, illumination-sensitive, and difficult to localize precisely**.

Harris Corner Detection identifies **stable and distinctive points where intensity changes in two directions**. These points provide better localization, more reliable matching, and improved robustness against illumination and background variations.

Therefore, replacing edge detection with **Harris Corner Detection significantly improves the accuracy and reliability of satellite image registration**, making it more suitable for environmental monitoring applications.

2. Choosing Grayscale or Binary Images for Real-Time Object Detection

Introduction

An autonomous surveillance system continuously captures high-resolution color images to detect **vehicles and pedestrians**. Processing full RGB images requires more computational resources because every pixel contains three color channels.

To increase processing speed, the images can be converted into:

1. **Grayscale representation**
2. **Binary representation**

The choice of representation affects **computational efficiency, edge detection, feature extraction, and object detection accuracy**.

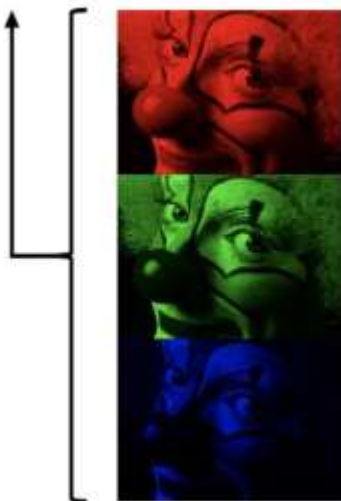
Color Images



Grayscale Images

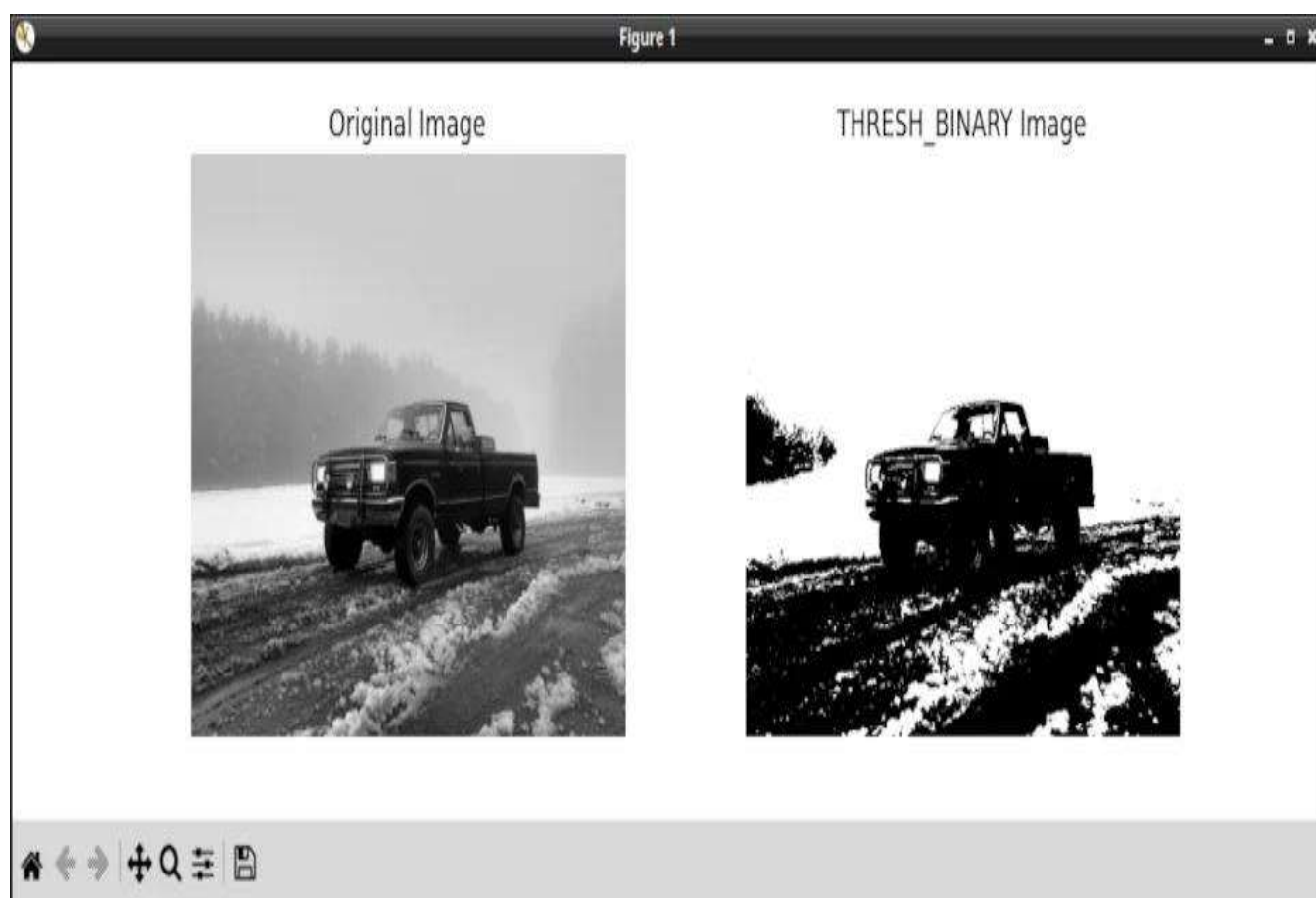
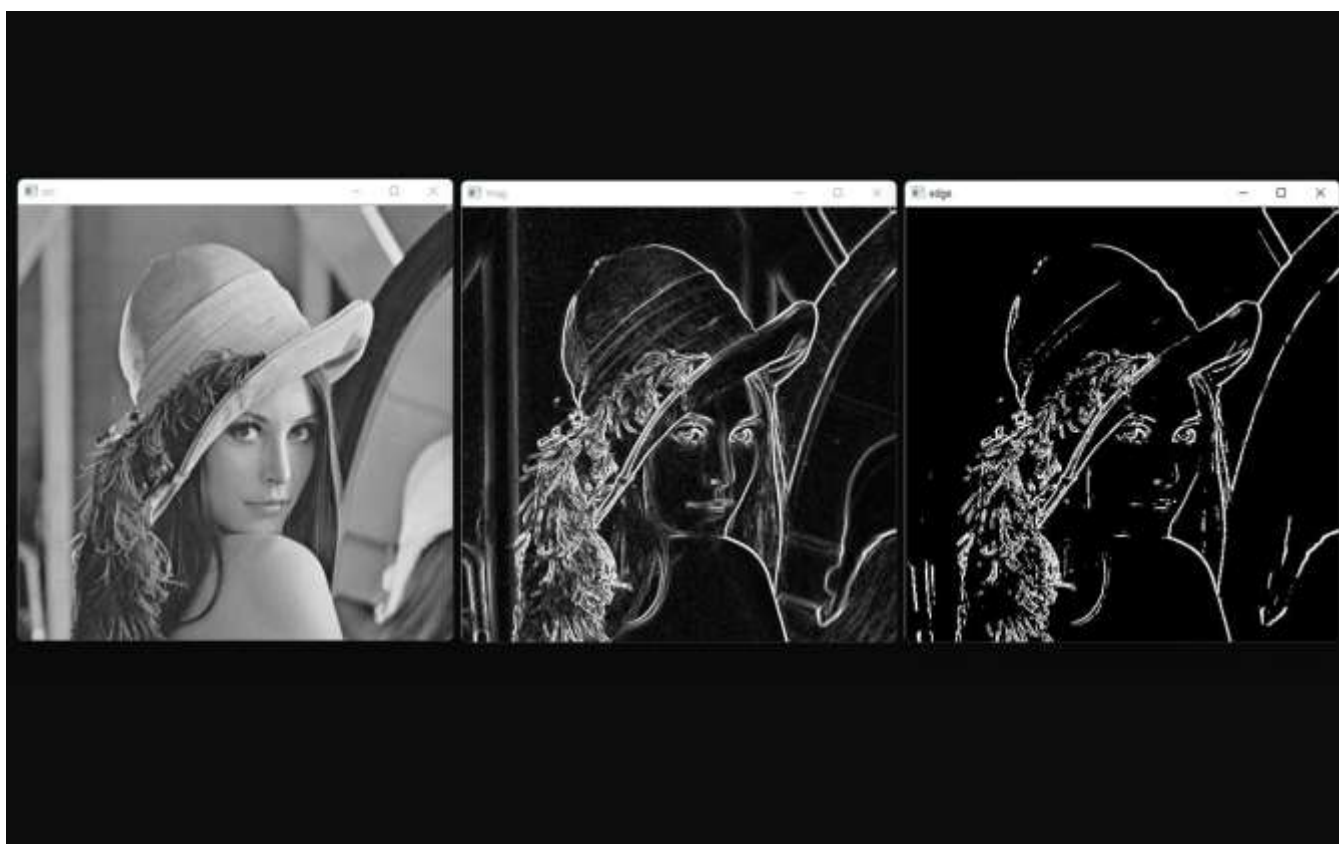


Binary Images



Used for most
scientific intensity
measurements

Used for masking/
segmentation



1. Color Image Representation

A color image normally contains three channels:

For an image of size :

pixel values must be processed.

This increases:

- memory usage
- processing time
- feature extraction complexity

However, color information can be useful for distinguishing objects.

For example, a red vehicle and a green background may be easily separated using color information.

2. Grayscale Representation

A grayscale image contains only one intensity channel.

Instead of:

each pixel has a single value:

Usually the intensity ranges from:

where:

- 0 = black
- 255 = white

A common conversion is:

Advantages

Reduced computational complexity

Only one channel needs to be processed instead of three.

Faster feature extraction

Algorithms such as:

- Harris Corner Detection
- SIFT
- Canny
- Sobel
- HOG

can work effectively with grayscale images.

Good edge detection

Edges are primarily detected using intensity changes.

For example, the Canny edge detector normally works on a grayscale image.

Reduced memory requirement

A grayscale image requires approximately one-third of the channel data of an RGB image.

3. Binary Representation

A binary image contains only two values:

or commonly:

For example:

Original image



Grayscale



Thresholding



Binary image

Thresholding can be represented as:

where T is the threshold.

Advantages

Binary images are extremely simple to process.

They can be useful for:

- object segmentation
- shape analysis
- contour detection
- connected component analysis

They require very little computational processing.

Problems with Binary Representation

Although binary images are computationally efficient, they lose a large amount of information.

1. Loss of intensity information

A grayscale image contains 256 possible intensity levels, whereas a binary image contains only two.

Therefore:

much information is lost.

2. Poor performance under changing illumination

Suppose a pedestrian is captured under:

- sunlight
- shadow
- nighttime lighting

A fixed threshold may incorrectly classify parts of the person as background.

3. Loss of object details

Small details such as:

- clothing patterns
- vehicle structure
- body contours
- facial/shape details

may disappear after thresholding.

4. Incorrect segmentation

If the vehicle and background have similar intensity values, thresholding may fail to separate them.

Therefore, binary representation is not always reliable for real-world surveillance.

Comparison of Image Representations

Property	RGB	Grayscale	Binary
Channels	3	1	1
Computational cost	High	Low	Very low
Memory usage	High	Lower	Very low
Intensity information	High	Moderate	Very low
Edge detection	Good	Excellent	Depends on threshold
Feature extraction	High cost	Efficient	Limited
Object details	Excellent	Good	Poor
Illumination robustness	Better in some cases	Moderate	Low
Real-time processing	Expensive	Very suitable	Very fast
General object detection	Excellent	Very good	Limited

Most Appropriate Representation

For a **real-time surveillance system**, **grayscale representation is the best general choice**.

The processing pipeline can be:

Color Image



Grayscale Conversion



Noise Reduction



Edge / Feature Detection



Object Detection



Vehicle / Pedestrian Recognition

Grayscale provides a good balance between **speed and accuracy**.

It significantly reduces computational complexity while preserving the intensity information required for edge and feature detection.

Binary images are faster but should generally be used **after segmentation or thresholding**, rather than as the primary representation for general vehicle and pedestrian detection.

Example

Suppose a surveillance camera captures:

RGB Image

$1920 \times 1080 \times 3$

After grayscale conversion:

Grayscale

$1920 \times 1080 \times 1$

The feature detector now processes only one intensity channel.

This reduces the amount of data while retaining important structures such as:

- object boundaries
- corners
- shapes
- contours
- textures

Therefore, algorithms such as **Harris, SIFT, HOG, Sobel, and Canny** can operate efficiently.

Conclusion

RGB images preserve the maximum amount of information but require greater computational resources. Binary images provide the highest processing speed but lose significant image information and are sensitive to threshold selection.

Grayscale is the most appropriate representation for real-time vehicle and pedestrian detection because it provides an effective compromise between computational efficiency and detection accuracy. It reduces processing requirements while retaining sufficient intensity information for reliable edge detection and feature extraction.